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ON FATES COMPARABLE TO DEATH*

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Earlier models for the evaluation of assuming and shedding risks of death are extended to the case of possibly extreme disability over a lifetime. Earlier micromort values are supplemented with microdisability values to enable individuals to make consistent decisions in hazardous situations.

(RISK ASSESSMENT; LIFE VALUE; DISABILITY)

Introduction

Two fallacies have often impeded the study of decision-making about hazards to human life. The first fallacy—the altruistic fallacy—is that the value a person places on his life is primarily related to such measures as the life insurance he carries, court judgments, and discounted future earnings—even though these measure mainly the value of his life to others rather than to himself. The effect of this fallacy is usually to place value on life too low for people to accept in making their own decisions.

The second fallacy is the incremental fallacy. Simply stated, it is the belief that because a person is willing to incur additional risk for money, one can infer a cash payment that the person should accept for being killed. Clearly, this is absurd, but this fallacy is often why discussions about placing values on lives have seemed both silly and frightening to the public.

The price a person would exact to accept a given incremental risk should rise as the total risk the person faces increases. No conceptualization of life decision-making is complete until it has captured this phenomenon.

This paper extends previous work concerned with risks to life (Howard 1978, 1979, 1980, 1981, Howard et al. 1978) to include the possibility of living with various degrees of disability. We shall investigate not only risks of disability themselves, but also the interaction between death and disability risks that such events as medical operations might pose.

The Ethical Foundation and Its Implications

Before proceeding, we must be clear on the ethical framework of the analysis. Our ethical assumption is that each person, and only that person, has the right to make or to delegate decisions about risks to his life or well-being. We note in passing that several present social arrangements do not meet this ethic and are the source of continuing controversy; for example, consider the development of the “right-to-die” movement.

Once we accept this ethic and deal with an individual who is rational in that he desires to follow the axioms of probabilistic logic, we can develop a procedure that will allow the individual to make consistent rational choices about the hazards in his life. In particular, the procedure focuses on what the person’s life is worth to himself, using his preferences for different life states. Furthermore, it shows that an individual is consistent in being willing to trade risk of life or disability for money at low risk levels and yet refusing to do so at high risk levels.

* Accepted by Rakesh K. Sarin, Special Editor.

Summary of Previous Work

We shall now summarize briefly the model we will use and extend in this paper. Readers are referred to other sources (Howard 1979, 1980) for a more complete description. In the simplest model investigated, we assume the individual has a fundamental preference on consumption and length of life and ignore the question of legacies. Consumption is defined as the constant level of consumption beyond bare survival over the remainder of life, measured in real dollars, that would be equivalent to current expectations; this is called the constant annual consumption c . When we combine c with the remaining years of life l , we have the fundamental descriptors of life quality and quantity. We assume that the individual can trade between c and l to develop a worth numeraire $w(c, l)$, and we use the simple model: $w(c, l) = c(l/\bar{l})^\eta$, $\eta \geq 0$. Here η is the parameter governing the trade-off, and $\bar{l}$ is the expected lifetime remaining, a useful benchmark, when $l = \bar{l}$, $w(c, l) = c$.

In fact, the emphasis of this model is on percentage changes in c and l rather than on their magnitudes. When l is small, this property may not be appropriate, and other models may be useful (Howard 1979).

Now we must specify the risk preference of the individual on the numeraire w . We use the exponential form with risk tolerance ρ , $u(w) = -e^{-w/\rho}$.

To find the payments that will compensate for risk, we use those payments to modify c in this model. We assume that any payment received will be used to buy an annuity at the prevailing interest rate i from a seller who agrees with the buyer's probability distribution on life. Payments to be made can be financed in a complementary manner. For example, if the individual receives one dollar, then he will be able to increase his annual consumption by

$$\zeta = \frac{1}{1+i} \frac{1}{1 - \langle (1/(1+i))^l \rangle}$$

where $\langle \rangle$ denotes expectation.

One result is immediately derivable from this model. If the individual is offered larger and larger amounts of money to face a probability of imminent death p , there is a maximum value of p , $p_{\max} = \langle e^{-(c/p)(l/\bar{l})^\eta} \rangle$. No amount of money, however, would induce the individual to accept a risk of death as large as $p_{\max}$.

The Micromort Value

While curves can be, and have been, derived to determine the payments $x(p)$ required for risks of death $p < p_{\max}$, the case of small incremental risk is of special interest. We define v_s , the small-risk value of life, as

$$v_s = \lim_{p \rightarrow 0} \frac{x(p)}{p},$$

which is readily computable from the model. If a person had calculated his small-risk life value, then it would suffice for all decisions in the safety range, say for $p < 1/1000$. To determine the death risk to him, in dollars, that should be compared with other costs and benefits, a person would simply multiply p by v_s .

However, since even the small-risk life value may lead some to the incremental fallacy, it is better to use v_s in the form of the value of a micromort [μm], a 10^{-6} chance of death. Then $v_{\mu m} = 10^{-6}v_s$. Now, by keeping in mind a micromort value $v_{\mu m}$ when confronted with a risk in the safety range, any person can simply compute the number of micromorts in the risk, multiply by $v_{\mu m}$, and establish the death risk in dollars. Although this change is cosmetic only, we should remember the size of the cosmetic industry.

As we have said, the model described is the simplest one we have analyzed that both possesses the desired qualities of finite $p_{\max}$ and a small-risk life value and is rich enough to suit many tastes. However, to represent a wide range of value functions and risk preferences, more general formulations are available (Howard 1979).

We should note that the concept of developing a worth numeraire on attributes and then assessing a risk preference upon it is as general a procedure as using a multi-attribute utility function.

Numerical Results

To obtain a feeling for the model, it is useful to summarize previous numerical results. Consider a base-case individual—a 25-year-old male with a \$20,000 per year constant annual consumption. He chooses $\eta = 2$, which means he would forgo 2% of his consumption over his remaining life to have it be 1% longer. His risk tolerance is \$6000, which means, for example, that he is roughly indifferent between his present situation and a lottery with a 2/3 chance of \$32,000 per year and a 1/3 chance of \$14,000 per year. The prevailing interest rate i is 5%.

How much would this individual have to be paid to accept a probability of death p ? Figure 1 shows the answer. Note that no amount of money will induce him to accept a risk of death greater than $p_{\max} = 0.103$. In the safety region, the curve becomes a straight line corresponding to a small-risk life value of \$2.43 million.

If the individual can avoid a risk of death p by paying money, then we obtain the curve of Figure 2. Even to avoid certain death, he would not be able to pay more than the value of the annuity represented by his constant annual consumption c/ξ ; we call this the economic value of his life, $v_e = \$363,000$. The straight-line portion of the curve in the safety region again corresponds to a small-risk life value of \$2.43 million.

Thus, while there is a considerable difference between buying and selling risks for large probabilities of death, the treatment is symmetric in the safety region. The

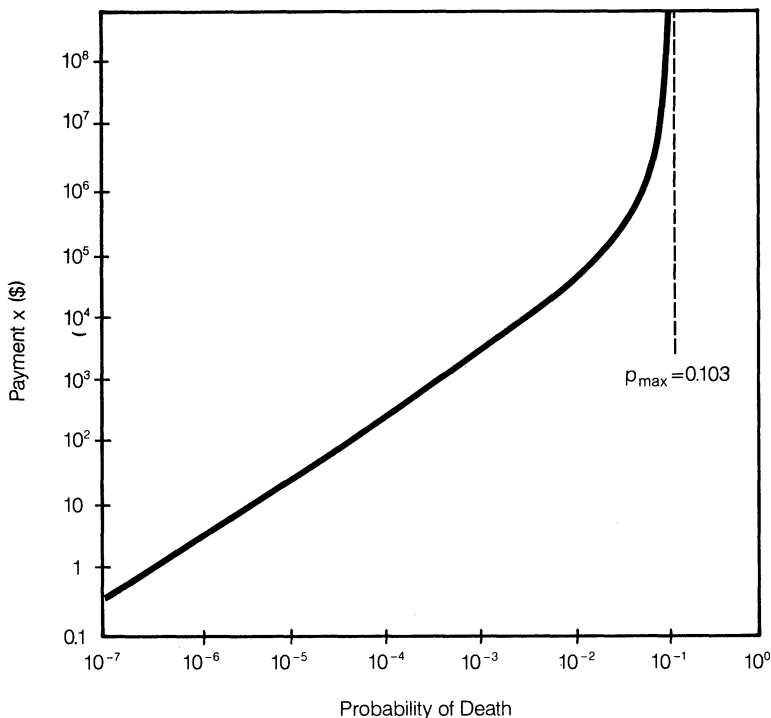


FIGURE 1. Amount That Base-Case Individual Would Have to Receive to Accept a Risk of Death p .

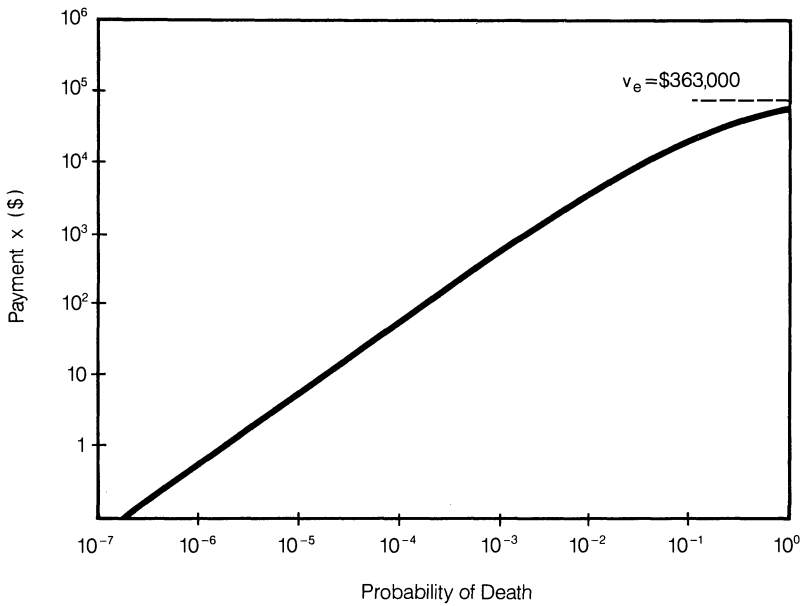


FIGURE 2. Amount That Base-Case Individual Would Pay to Avoid a Risk of Death p .

individual is willing to buy and sell micromorts at a price of \$2.43. If someone wanted him to take on a risk of 1/10,000 probability of death, he would evaluate the death risk at \$243 (100 μm). Of course, if someone wanted to buy 1,000,000 μm , he would refuse. This micromort value will increase as the individual takes on more and more risk; however, for most of us who are both buying and selling very small risks all the time, it should be relatively constant and hence a useful guide to safety decisions. Naturally, the death risk cost computed in this way must be combined with other costs and benefits to arrive at a decision.

Continuing Hazards

While we have discussed risk as if it will occur only in the present year, many of life's activities, like smoking, for example, imply a change in the risk of death in future years (Howard 1979, 1980). Knowing the changes in risks in future years will enable us to derive the corresponding lifetime distribution and to use the model to compare this situation with any other.

A serious hazard, like smoking, which might double the chances of dying in any year, would reduce life expectancy for the base-case individual from 46.2 to 38.4 years. To be consistent, he would demand a lump-sum payment of \$212,000 or an annuity of \$12,400 before assuming such a risk. If smoking is not worth that much to him, he should reconsider whether to pursue it.

Health States

Now let us show how to use this same structure with a slightly more complex description of life. We shall retain constant annual consumption as the economic indicator of life, but consider more possibilities for the noneconomic aspects of life. Suppose we define a number of health states $i = 1, 2, \dots, n$ for the individual that he might occupy at any time in his life. Such states might be defined in many ways; we are concerned here, however, with the concept. The states might be characterized by the potential mobility of the person, by the amount of pain he feels, or by the senses he has available. For any life, the transitions through the states would be probabilistic,

both as to the succession of states and as to the time spent in each. A semi-Markov model would be a fairly simple probabilistic model to describe this process, provided that the Markovian assumption was justified in this particular case (Howard 1971). For related approaches, see Fanshel and Bush (1970), and Shepard and Zeckhauser (1976).

To complete the description of the system, we need to extend the earlier value model, which depended only on c and l . As a simple extension, let us retain the original form for $w(c, l)$, but let l now be a weighted sum of the years spent in each state i , l_i , $l = \sum_i f_i l_i$. For the state of full health and capacities, the weight would be one; however, for less desirable states, it would be less than one.

For example, suppose a person faced a normal lifetime except that from this point on he would be blind. If he said that living his remaining life blind would be equivalent to living only 30 percent of that life sighted, then the weighting for the blind years would be 0.3. While more complicated schemes can be developed to represent preferences, let us see how valuable this simple model can be.

One question that immediately arises is whether the weights can be negative; that is, are some life states so bad that living them would be worse than being dead? Many people feel that there are, seeing total paralysis as one such state. Since we have assumed that each person has the right to make and delegate decisions about his own life, any person could choose suicide at any time and, thus, weight that life state as zero. The restriction to nonnegative weights is, therefore, not a problem for those who have suicide as an option.

Disability

We shall focus our analysis on the case of disability: where the person faces the possibility of spending the rest of his life with a serious health impairment, like blindness, paralysis, or severe pain. In each case, we imagine assessing the worth function by asking the person the following question: "Suppose that instead of living the rest of your life in the state of health you expect, you would live it in this way (a specified disabled state, like blindness or total paralysis). However, you have a choice—you can continue to live with the state of health you expect, but for a reduced time. What fraction of your remaining years lived in this state would be just equivalent to living all your remaining years in the disabled state?" The fraction f that the person answers will be used as the weighting factor in the value model. We can then see how the person would make various decisions involving a risk of disability or a risk of either death or disability. We shall assume that the fraction f is assessed independent of l , although there is only modest difficulty in making it a function of l . For the moment, we shall further assume that the person's income will be unaffected by his disability and, hence, concentrate on the qualitative aspects of the disabled state.

The Disability Black Pill

We proceed now in a manner analogous to our earlier work on death risk. Suppose the person is offered an amount of money x to assume a risk p of being disabled to a level f . For example, someone could be offered \$1000 to take a 1/10,000 chance of becoming totally paralyzed, an outcome he regards as equivalent to $f = 0.1$. The choice is diagrammed in Figure 3. For simplicity, we can imagine the risk is contained in a pill and that everyone agrees on the probability p . If the person refuses to take the pill, then he lives his normal life, receiving any windfalls and calamities that may be in store for him—his future life lottery. This has a utility of $\langle u(c, l) \rangle$, obtained by multiplying the utility of each constant annual consumption level c and remaining life l by the joint probability distribution on these quantities. If the person takes the pill, then with probability $1 - p$ he will live his future life lottery as before, only with more

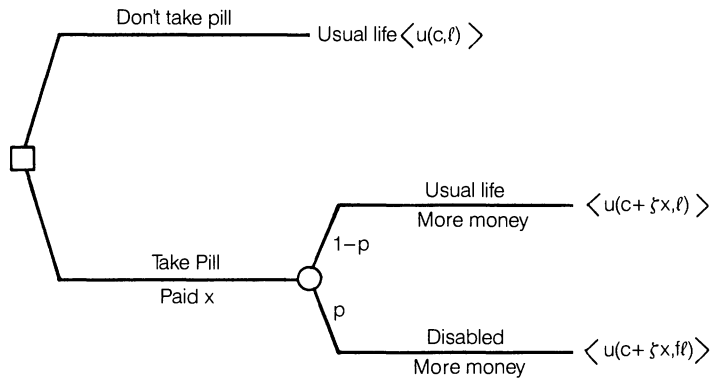


FIGURE 3. Disability Black Pill Tree.

money. He converts the payment x into the constant annual consumption $c + \zeta x$, with expected utility $\langle u(c + \zeta x, l) \rangle$. However, if he becomes disabled, which he will with probability p , then he transforms the payment in the same way, but has his remaining years multiplied by f so far as his value function is concerned. Therefore, he has expected utility $\langle u(c + \zeta x, fl) \rangle$. Setting the expected utility of the two alternatives equal, we find

$$\begin{aligned} \langle u(c, l) \rangle &= p \langle u(c + \zeta x, fl) \rangle + (1 - p) \langle u(c + \zeta x, l) \rangle \quad \text{or} \\ p &= \frac{\langle u(c + \zeta x, l) \rangle - \langle u(c, l) \rangle}{\langle u(c + \zeta x, l) \rangle - \langle u(c + \zeta x, fl) \rangle} . \end{aligned}$$

Base-Case Results

This equation allows us to calculate the risk of a disability level f that would be assumed for a payment x once all the other parameters of the model have been specified. With the results shown in Figure 4, we evaluate the equation for the

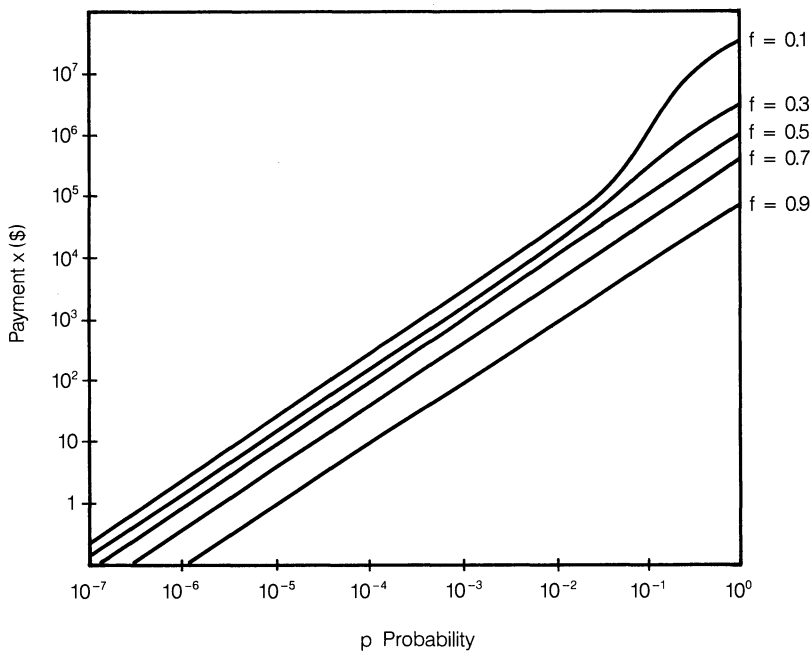


FIGURE 4. Disability Black Pill Results.

TABLE I
Small-Risk Value of Disability at Level f for Base-Case Individual

f	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$v_{sd}(f)$ [M\$]	2.333	2.066	1.696	1.297	0.927	0.620	0.383	0.210	0.087
$v_{\mu d}(f)$ [\$]	2.33	2.07	1.70	1.30	0.93	0.62	0.38	0.21	0.09

base-case individual described earlier. We see that the curves begin to diverge from straight lines on these scales only for p close to one and f small, that is, when there are severe risks of serious disabilities. Within the safety region ($p = 10^{-3}$ or less), they are all straight lines. This means that in this region we can define a small-risk value of disability level f , $v_{sd}(f)$ that can be used by the individual in the expected value sense to compute x as $x = pv_{sd}(f)$, where

$$v_{sd}(f) = \lim_{\substack{x \rightarrow 0 \\ p \rightarrow 0}} \frac{1}{dp/dx} .$$

If we define $v_{\mu d}(f) = 10^{-6}v_{sd}(f)$, then we have the value of a 10^{-6} chance of a disability level f in consonance with our definition of a micromort. The results for the base-case individual are listed in Table I and plotted in Figure 5. Notice that the microdisability value approaches the micromort value of \$2.43 when f is small, and that it approaches zero as f approaches 1.

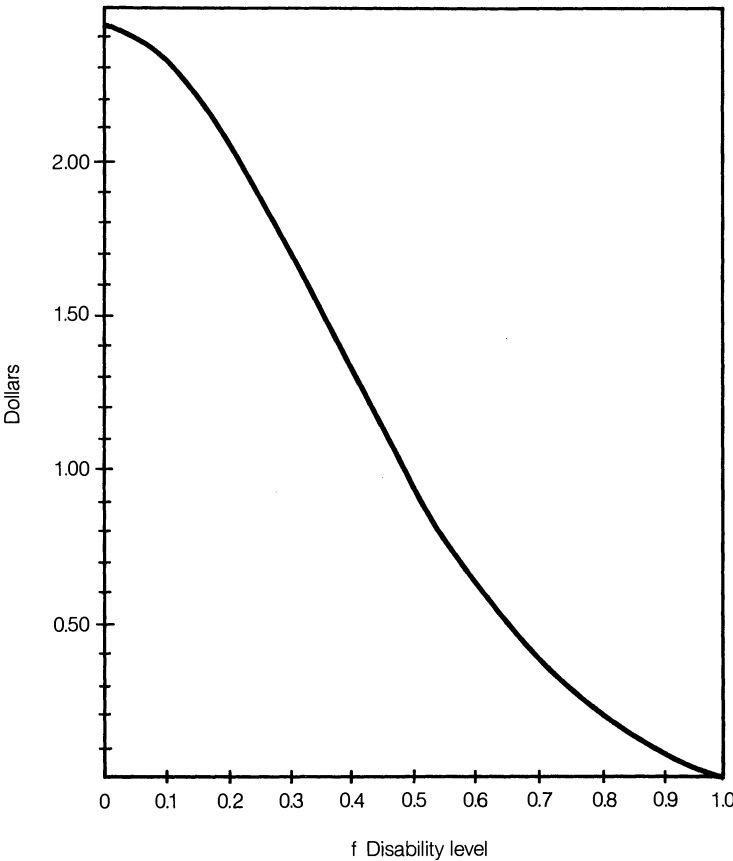


FIGURE 5. Value of 10^{-6} Chance of Disability to Level f for Base-Case Individual $v_{\mu d}(f)$.

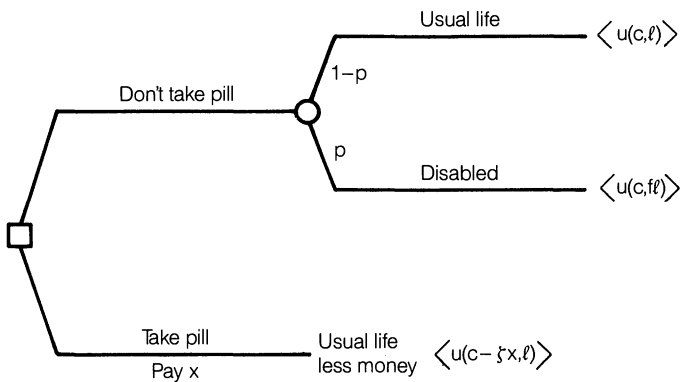


FIGURE 6. Disability White Pill Tree.

The Disability White Pill

We can also think of selling risks of being disabled. Suppose that a person facing a risk p of being disabled to level f could avoid that risk for a payment x . How much should he pay? We call this the disability white pill question and diagram it as shown in Figure 6. Note that if the person pays x , he finances the payment by reducing his constant annual consumption to $c - \xi x$. Of course, the most he could pay would be $x = c/\xi = v_e$, the economic value of his life. Equating the utility of the two alternatives, we find

$$\begin{aligned} \langle u(c - \xi x, \ell) \rangle &= p \langle u(c, \ell f) \rangle + (1 - p) \langle u(c, \ell) \rangle \quad \text{and} \\ p &= \frac{\langle u(c, \ell) \rangle - \langle u(c - \xi x, \ell) \rangle}{\langle u(c, \ell) \rangle - \langle u(c, \ell f) \rangle} . \end{aligned}$$

Base-Case Results

Figure 7 shows how much the base-case individual would pay to avoid a probability p of a disability level f . Note that the curves are again straight lines in this plot until

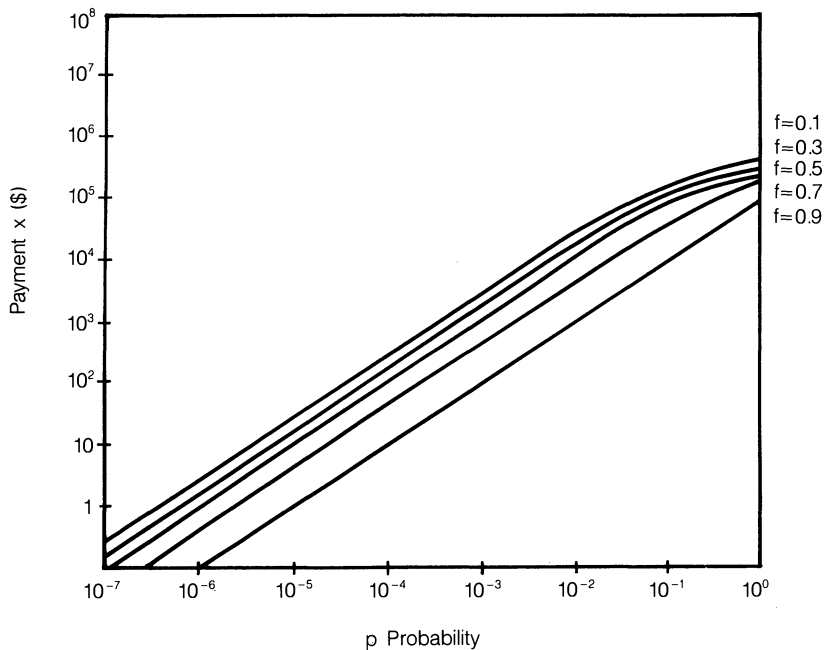


FIGURE 7. Disability White Pill Results.

they approach $p = 1$. In this region, they bend downward as the individual encounters the finiteness of his resources. However, for values of p in the safety region, they are straight lines and, in fact, the same straight lines as in Figure 4. This means that the individual will buy and sell small risks of being disabled for the microdisability values in Table I and Figure 5. He thus treats small risks of disability in the same way as small risks of death.

Changes in Disability Level

We can also consider the risks of changing from one level of disability to another. In particular, we shall consider the case of a person who is currently disabled becoming more disabled, even to the point of death. The situation is diagrammed in Figure 8: a person who is currently disabled at level f_1 is offered x to assume a probability p of being disabled to a more restricted level f_2 , including even the possibility of death, $f_2 = 0$. To find the x to which he is indifferent, we equate the utility of the two alternatives,

$$\langle u(c, f_1 l) \rangle = p \langle u(c + \xi x, f_2 l) \rangle + (1 - p) \langle u(c + \xi x, f_1 l) \rangle$$

and find

$$p = \frac{\langle u(c + \xi x, f_1 l) \rangle - \langle u(c, f_1 l) \rangle}{\langle u(c + \xi x, f_1 l) \rangle - \langle u(c + \xi x, f_2 l) \rangle}$$

where $1 \geq f_1 > f_2 \geq 0$. If we let $v_{sd}(f_1, f_2)$ be the small-risk value of disability from level f_1 to level f_2 defined by

$$v_{sd}(f_1, f_2) = \lim_{\substack{x \rightarrow 0 \\ p \rightarrow 0}} \frac{1}{dp/dx},$$

then in the safety region we can compute the required payment x from $x = pv_{sd}(f_1, f_2)$ or from the equivalent microdisability value of this change

$$v_{\mu d}(f_1, f_2) = 10^{-6} v_{sd}(f_1, f_2).$$

The results for the base-case individual are shown in Table II. Notice that the first row of the table reproduces the small-risk values of Table I for the case of risk of disability to a healthy individual. In general, the value decreases as f_1 decreases for a fixed f_2 and increases as f_2 decreases for a fixed f_1 . The first phenomenon results from contemplating the same prospect from worsening initial states, and the second results from contemplating worsening final states from the same initial state.

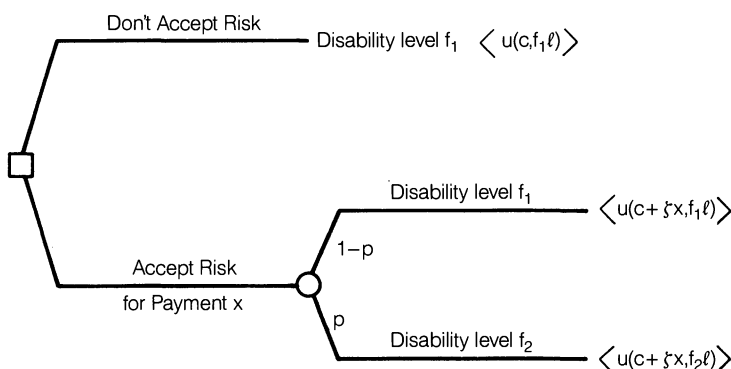


FIGURE 8. Change in Disability Level Tree.

TABLE II
Small-Risk Value of Changing from Disability Level f_1 to Disability Level f_2 , $f_1 > f_2$
(Thousands of Dollars)

From f_1 \ To f_2	0.9	0.8	0.7	0.6	0.5	0.4	0.3	0.2	0.1	0
1.0	87	210	383	620	927	1297	1696	2066	2333	2431
0.9		96	231	416	656	944	1256	1547	1752	1829
0.8			107	254	444	673	920	1149	1314	1374
0.7				120	277	464	667	855	991	1041
0.6					136	300	477	641	759	802
0.5						156	325	482	594	636
0.4							184	354	477	522
0.3								225	386	446
0.2									291	398
0.1										372

Combined Hazards of Disability and Death

Deciding to participate in certain activities, like driving a motor vehicle or operating a chain saw, entails hazards ranging from minor disabilities through death. Suppose there are n such hazards, and let p_i be the probability that the individual will be disabled to level f_i , $i = 1, 2, \dots, n$, where $f_i = 0$ represents death. Then, the total probability of one of these outcomes is $p = \sum_{i=1}^n p_i$, and the conditional probability of disability to level i given that some hazard occurs is $q_i = p_i/p$. The amount x that the individual would have to be paid to take on this combined hazard can be determined from Figure 9 by equating the expected utility of the two alternatives:

$$\langle u(c, l) \rangle = (1 - p) \langle u(c + \zeta x, l) \rangle + p \sum_{i=1}^n q_i \langle u(c + \zeta x, f_i l) \rangle \quad \text{or}$$
$$p = \frac{\langle u(c + \zeta x, l) \rangle - \langle u(c, l) \rangle}{\langle u(c + \zeta x, l) \rangle - \sum_{i=1}^n q_i \langle u(c + \zeta x, f_i l) \rangle}.$$

We can now let v_{sh} be the small-risk value of the combined hazard, defined by

$$v_{sh} = \lim_{\substack{p \rightarrow 0 \\ x \rightarrow 0}} \frac{1}{dp/dx}$$

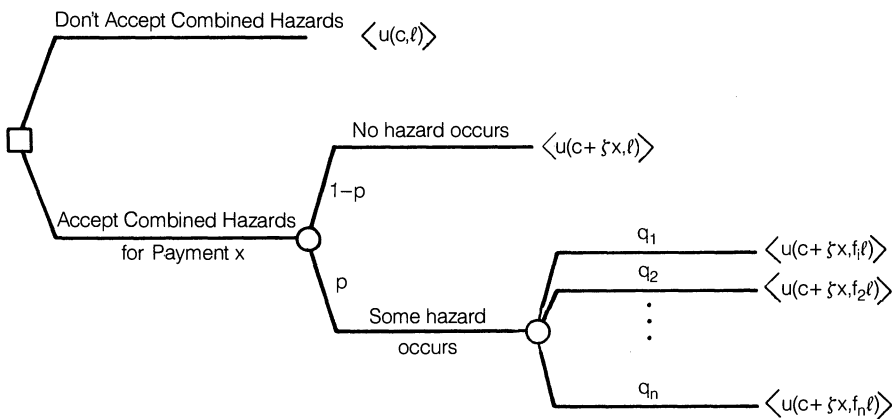


FIGURE 9. Combined Hazard Tree.

and compute x using $x = pv_{sh}$ when the probability p is in the safety region. Equivalently, we can use $v_{\mu h} = 10^{-6}v_{sh}$ as the microhazard value of a 10^{-6} chance of combined hazard.

To illustrate these computations, suppose that the base-case individual confronts a situation where he has probabilities $p_1 = 0.00005$ of death, $p_2 = 0.00025$ of disability level $f_2 = 0.3$ (perhaps from losing a leg), and $p_3 = 0.0007$ of disability level $f_3 = 0.9$ (perhaps from losing a finger). We find that the microhazard value is \$0.606 and, since $p = 1000 \times 10^{-6}$, that the required payment to accept this combined hazard would be $x = \$606$.

An equivalent and more convenient way to compute x in the safety region is to add the amounts the base-case individual would have to be paid to assume each of the individual hazards, $x = \sum_{i=1}^n p_i v_{sd}(f_i)$ where $v_{sd}(0)$ is equal to v_s , the small-risk value of life. If p_i is measured in microprobability (10^{-6}) units, then $x = \sum_{i=1}^n p_i v_{ud}(f)$, and using the values in Table I, we find

$$\begin{aligned} x &= 50(2.430) + 250(1.696) + 700(0.087) \\ &= 121.5 \quad + 424 \quad + 60.9 \\ &= 606. \end{aligned}$$

We see that the major contributor to the payment is the \$424 required by the threat of disability to level 0.3.

The payment to accept combined hazards in the safety region is thus easily computed by adding the payments required to accept each of them individually.

Disability or Death

Suppose an individual disabled at level f faces an operation that may cure him at some risk p of death. How large a risk of death could he tolerate as a function of his disability level? The choice is diagrammed in Figure 10. Equating expected utilities produces

$$p = \frac{\langle u(c, l) \rangle - \langle u(c, fl) \rangle}{\langle u(c, l) \rangle - \langle u(0, 0) \rangle}.$$

We can, therefore, determine the maximum death risk p that would be associated with each level of disability f . The results for the base-case individual appear in Figure 11. Note that a severely handicapped individual could tolerate very high death probabilities. This corresponds to the fact that the riskiest treatments are reserved for patients without much to lose.

We can easily add the possibility that the operation will be costly as well as

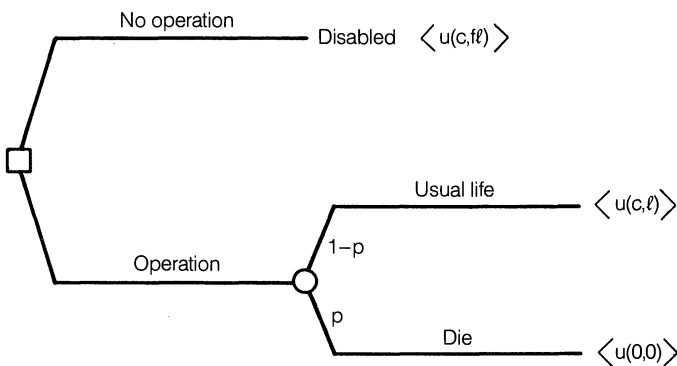


FIGURE 10. Operation Decision Tree.

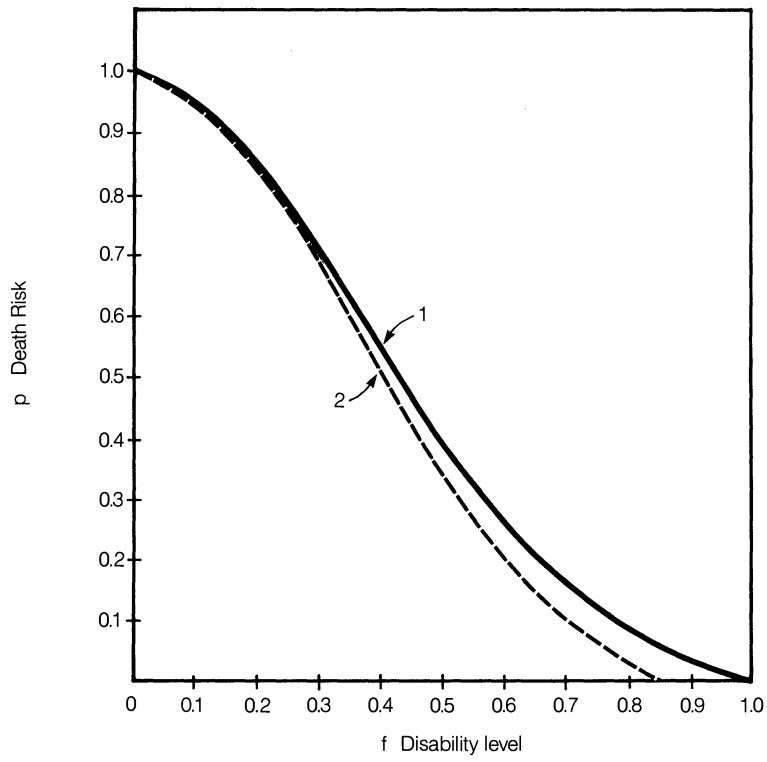


FIGURE 11. Maximum Probability of Death from Operation That Could Be Tolerated by Base-Case Individual with Disability Level f .
1. For free operation.
2. For \$100,000 operation.

dangerous. Figure 12 diagrams the case where the operation costs the patient an amount x . We note that even if $p = 0$, the operation may not be desirable. If $\langle u(c, fl) \rangle < \langle u(c - \xi x, l) \rangle$, then the operation costs so much that the patient considers himself worse off than he is now even if it succeeds. The maximum payment for the operation is obtained by setting the two utilities equal

$$\begin{aligned} \langle u(c, fl) \rangle &= \langle u(c - \xi x, l) \rangle, & w(c, fl) &= w(c - \xi x, l), \\ c \left(\frac{fl}{l} \right)^\eta &= (c - \xi x) \left(\frac{l}{l} \right)^\eta, & \text{or} \\ x &= \frac{c}{\xi} [1 - f^\eta] = v_e [1 - f^\eta]. \end{aligned}$$

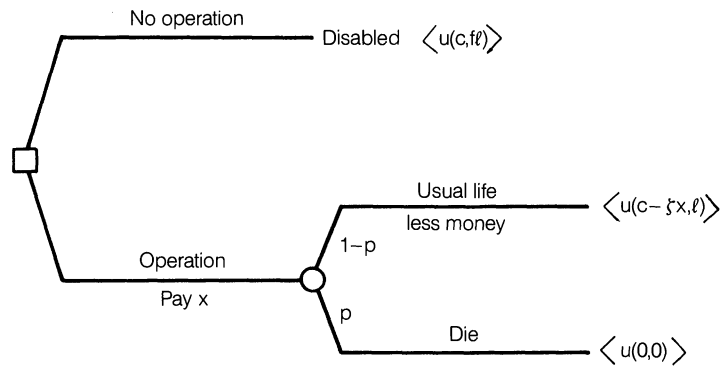


FIGURE 12. Costly Operation Decision Tree.

TABLE III
Maximum Payment for Operation as Function of Disability Level

Disability Level	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Payment [K\$]	363	359	349	330	305	272	232	185	131	69	0

If $f = 0$, the individual would pay his whole economic value v_e , which is \$363,048. As f increases, the amount he would pay falls as shown in Table III.

Provided we limit the payment as discussed, we can then use the tree of Figure 12 to write

$$p = \frac{\langle u(c - \zeta x, l) \rangle - \langle u(c, fl) \rangle}{\langle u(c - \zeta x, l) \rangle - \langle u(0, 0) \rangle} ,$$

and compute the maximum tolerable probability of death for an individual at disability level f facing an operation that costs x .

We have done this for the base-case individual for the cost $x = \$100,000$ —the dashed line in Figure 11. For seriously disabled people, the cost is not important, but for moderately disabled people, it is. No operation would be considered at a disability level of 0.85. At $f = 0.7$ only a 0.10 rather than a 0.16 probability of death could be tolerated.

Compensation

We can use the model even without risk to determine the amount of compensation that an individual would require to be indifferent to a given disability level. Thus, to determine the amount x he would have to be paid for being placed at disability level f , we equate

$$\begin{aligned} \langle u(c + \zeta x, fl) \rangle &= \langle u(c, l) \rangle, \\ w(c + \zeta x, fl) &= w(c, l), \\ (c + \zeta x) \left(\frac{fl}{l} \right)^\eta &= c \left(\frac{l}{l} \right)^\eta, \quad \text{or} \quad x = \frac{c}{\zeta} \left(\frac{1}{f^\eta} - 1 \right). \end{aligned}$$

The annual consumption increase required would be

$$\zeta x = c(1/f^\eta - 1).$$

Table IV shows the results for the base-case individual. Given that his constant annual consumption is \$20,000, that amount would have to be doubled to make him indifferent to a 0.7 level of disability.

TABLE IV
Compensation Required by the Base-Case Individual for a Disability Level f

Disability Level f	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Lump-Sum Compensation x [K\$]	35,942	8,713	3,671	1,906	1,089	645	378	204	85	0
Annual Payment ζx [K\$]	1,980	480	202	105	60	36	21	11	5	0

Income Effects of Disability

Up to this point, we have assumed that the effect of disability is to decrease the desirability of the life experience, but not to change either the life expectancy or the income received by the person. Let us now relax this income assumption, since disability will usually depress income. A simple way to investigate this effect is to assume that a person's income will be reduced by the same factor f that appears in his value function. (Naturally, we could make the reduction factor different from f if necessary.) The net effect of this assumption is that in every expression we have seen where we have multiplied l by f we now also multiply c by f .

Consider, for example, the effect of this change on the microdisability value $v_{\mu d}(f)$. Figure 13 shows that when this income effect is incorporated, microdisability values for the base-case individual increase substantially at every level of f except towards $f = 0$ and $f = 1$ where they are bounded respectively by the micromort value and by zero. At $f = 0.5$, the increase is from \$0.927 M to \$1.484 M. The prospect of losing both health and income is significantly more worrisome than losing health alone.

As we would expect, including the income effect makes the base-case individual more willing to accept a potentially fatal operation that could cure him. Figure 14 shows an increase in the tolerable death risk at all levels except $f = 0$ and $f = 1$. At $f = 0.5$, for example, the tolerable death risk for a free operation increases from 0.38 to 0.61.

Just as we have been able to explore the income effect rather easily within the model, so too could we incorporate the combined effects of disability on health state, income, and life expectancy.

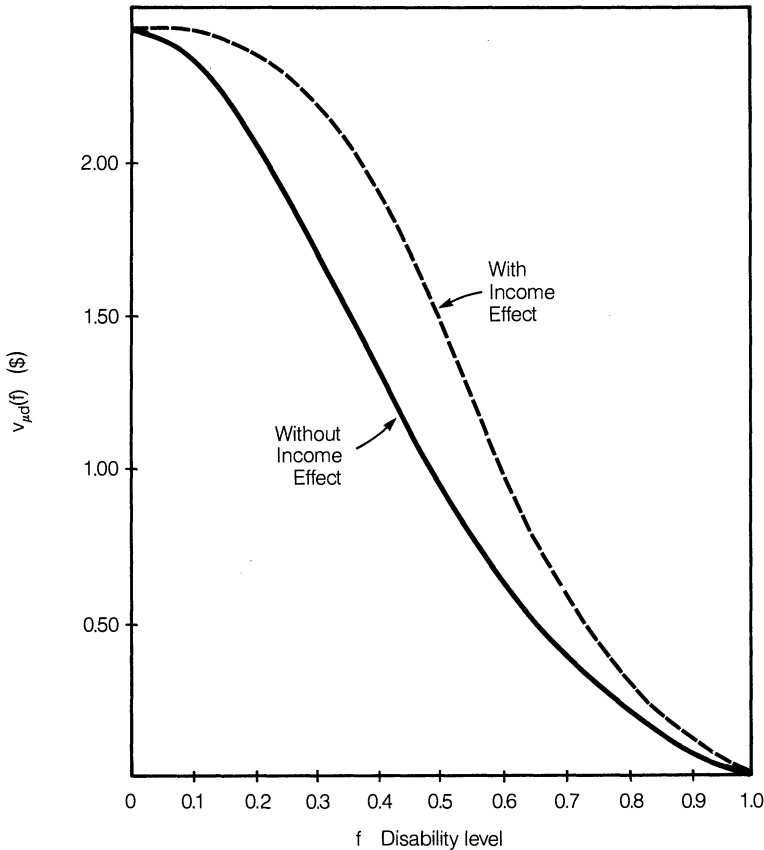


FIGURE 13. Increase in Microdisability Value Caused by Income Effect.

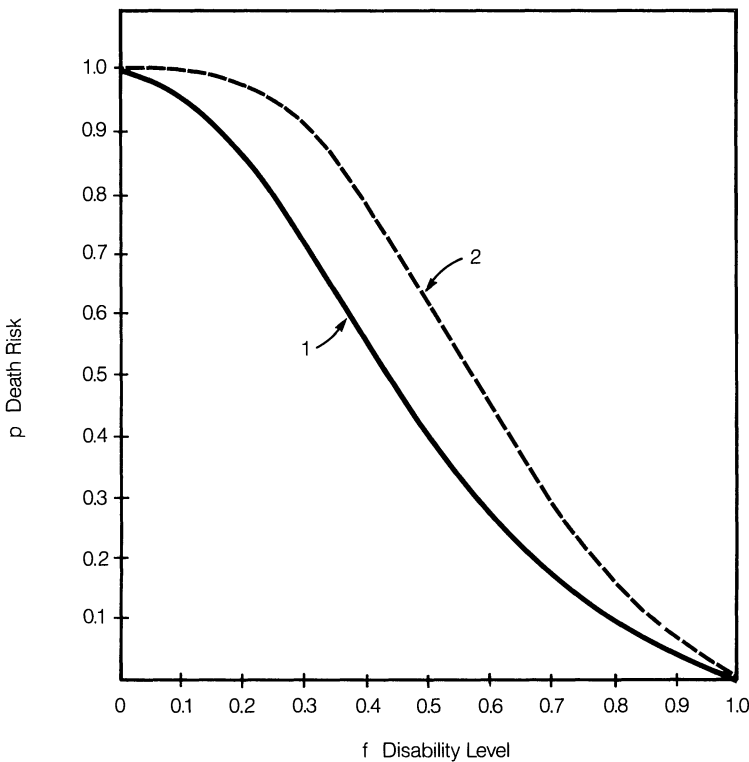


FIGURE 14. Maximum Probability of Death from Operation That Could Be Tolerated by Base-Case Individual with Disability Level f .

1. For free operation.
2. For free operation with income effect.

Conclusion

Analyzing fates comparable to death has required only a relatively straightforward extension of the models used earlier to analyze risks of death. The idea of a small-risk value that can be used to evaluate safety decisions is directly applicable to the case of disability. Using the combined results of these analyses, an individual can evaluate safety situations where he faces both risks of death and disability.

The concept of micromort, microdisability, and microhazard values is more important than the details of the model used to derive them. Even without an analytic model, an individual could directly assign such values and then use them for decisions in the safety region. While he would not be assured of consistency with underlying preferences, he would be assured of consistency across different hazardous situations.¹

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References

- HOWARD, R. A., "Life and Death Decision Analysis," Second Lawrence Symposium on Systems and Decision Sciences, Berkeley, California, October 1978.
- , "Life and Death Decision Analysis," Research Report EES DA-79-2, Department of Engineering-Economic Systems, Stanford University, December 1979.
- , "On Making Life and Death Decisions," *Societal Risk Assessment, How Safe Is Safe Enough?*, Richard C. Schwing and Walter A. Albers, Jr., (Eds.), General Motors Research Laboratories, Plenum Press, New York, 1980.
- , "The Risks of Benefit-Cost-Risk Analysis," *Risk / Benefit Analysis in Water Resources Planning and Management*, Yacov Y. Haimes (Ed.), Plenum Press, New York, 1981.
- , *Dynamic Probabilistic Systems*, Vols. I and II, John F. Wiley & Sons, Inc., New York, 1971.
- , J. E. MATHESON AND D. L. OWEN, "The Value of Life and Nuclear Design," Proceedings of the

- American Nuclear Society Topical Meeting on Probabilistic Analysis of Nuclear Reactor Safety, American Nuclear Society, May 8–10, 1978.
- FANSHEL, S. AND J. W. BUSH, "A Health Status Index and Its Application to Health Services Outcomes," *Oper. Res.*, 18 (1970), 1021–1066.
- SHEPARD, D. AND R. ZECKHAUSER, "Where Now for Saving Lives," *Law and Contemporary Problems*, (Fall 1976), 5–45.
- BODILY, S. E., "Analysis of Risks to Life and Limb," *Oper. Res.*, 28 (1980), 156–175.
- JONES-LEE, M., "The Value of Changes in the Probability of Death or Injury," *J. Political Econom.*, 82 (1974), 835–849.
- , *The Value of Life*, University of Chicago Press, Chicago, 1976.
- KEENEY, R. L., "Evaluating Alternatives Involving Potential Fatalities," *Oper. Res.*, 28 (1980), 188–205.
- LINNEROOTH, J., "The Evaluation of Public Programs Affecting Population Mortality," unpublished Ph.D. thesis, Department of Economics, University of Maryland, College Park, 1977.